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1 Low-rank channel mixing: where the factor r enters (and why Picard is unchanged)

This note isolates the only extra bookkeeping induced by the channel-mixing / low-rank structure

$$H_2(t, c_2; X, W) = \sum_{k=1}^{r} L_{c_2,k}\, m_k(t; X, W), \qquad m_k(t; X, W) = \mathbb{E}_{C_1}[w_1(t, C_1, k)\, \varphi_1(\langle f_1(C_1),\, X\rangle)]\,,$$

relative to the standard mean-field ODE well-posedness template (a priori ψ_2 bounds, good/bad event decomposition, and Picard iteration).

1.1 A deterministic r-factor for low-rank sums

It is convenient to record row norms of L:

$$\|L\|_{\infty,1} \equiv \sup_{c_2\in\Omega_2} \sum_{k=1}^{r} |L_{c_2,k}|, \qquad \|L\|_{\infty,2} \equiv \sup_{c_2\in\Omega_2} \Big(\sum_{k=1}^{r} L_{c_2,k}^2\Big)^{1/2}.$$

Under the entrywise bound $\sup_{c_2,k} |L_{c_2,k}| \le K$ (as in Assumption **??**),

$$\|L\|_{\infty,1} \le rK, \qquad \|L\|_{\infty,2} \le \sqrt{r}\, K.$$

Lemma 1 (Moment and ψ_2 control for low-rank sums)**.** *Let $(U_k)_{k=1}^r$ be real random variables on a common probability space and $(a_k)_{k=1}^r \in \mathbb{R}^r$ deterministic. Then for every $m \ge 1$,*

$$\mathbb{E}\Big[\big|\sum_{k=1}^{r} a_k U_k\big|^m\Big]^{1/m} \le \sum_{k=1}^{r} |a_k|\, \mathbb{E}[|U_k|^m]^{1/m} \le \Big(\sum_{k=1}^{r} |a_k|\Big) \max_{1\le k\le r} \mathbb{E}[|U_k|^m]^{1/m}.$$

Consequently, any "ψ_2-type" seminorm defined by $\sup_{m\ge1} m^{-1/2}\mathbb{E}[|\cdot|^m]^{1/m}$ satisfies

$$\sup_{m\ge1} \frac{1}{\sqrt{m}}\, \mathbb{E}\Big[\big|\sum_{k=1}^{r} a_k U_k\big|^m\Big]^{1/m} \le \Big(\sum_{k=1}^{r} |a_k|\Big) \max_{1\le k\le r} \sup_{m\ge1} \frac{1}{\sqrt{m}}\, \mathbb{E}[|U_k|^m]^{1/m}.$$

If one prefers an ℓ_2 version, then also $\sum_{k=1}^r |a_k| \le \sqrt{r}\,(\sum_k a_k^2)^{1/2}$ yields a $\sqrt{r}$ factor.

1.2 Application to H_2: $\|L\|_{\infty,1}$ is the only new constant

Lemma 2 (Channel mixing only multiplies forward bounds by $\|L\|_{\infty,1}$)**.** *Assume φ_1 is bounded by K. Then for every $t \ge 0$ and every $c_2 \in \Omega_2$,*

$$\sup_{s\le t} \big|H_2(s, c_2; X, W)\big| \le \Big(\sum_{k=1}^{r} |L_{c_2,k}|\Big) \max_{1\le k\le r} \sup_{s\le t} |m_k(s; X, W)| \le \|L\|_{\infty,1} \max_{1\le k\le r} \sup_{s\le t} |m_k(s; X, W)|.$$

Moreover,

$$\sup_{s\le t} |m_k(s; X, W)| \le \mathbb{E}_{C_1}\Big[\sup_{s\le t} |w_1(s, C_1, k)|\, \big|\varphi_1(\langle f_1(C_1),\, X\rangle)\big|\Big] \le K\, \mathbb{E}_{C_1}\Big[\sup_{s\le t} |w_1(s, C_1, k)|\Big]\,.$$

Combining these inequalities and taking moments yields, for every $m \geq 1$,

$$\mathbb{E}_X\Big[\sup_{s\leq t}\sup_{c_2}|H_2(s,c_2;X,W)|^m\Big]^{1/m} \leq K\,\|L\|_{\infty,1}\max_{1\leq k\leq r}\mathbb{E}_{C_1}\Big[\sup_{s\leq t}|w_1(s,C_1,k)|^m\Big]^{1/m}.$$

In particular, in the ψ_2-type calibration of Section 3,

$$\sqrt{50}\sup_{m\geq 1}\frac{1}{\sqrt{m}}\,\mathbb{E}_X\Big[\sup_{s\leq t}\sup_{c_2}|H_2(s,c_2;X,W)|^m\Big]^{1/m} \leq K\,\|L\|_{\infty,1}\,[\![w_1]\!]_{\psi,t}.$$

Remark 1 (The promised r factor)**.** *Under $\sup_{c_2,k}|L_{c_2,k}| \leq K$, Lemma 2 gives an explicit r-dependence*

$$[\![H_2]\!]_{\psi,t} \;\lesssim\; (rK)\,[\![w_1]\!]_{\psi,t},$$

up to the same universal constants already present in the two-layer well-posedness proof. If one instead assumes a row ℓ_2 control $\|L\|_{\infty,2} \leq K$ (e.g. normalized rows), then the same steps yield a $\sqrt{r}$ scaling.

1.3 Where this plugs into the Picard / well-posedness template

In the Picard map F (Section 4), the mixing kernel L appears only through H_2 and through the backpropagated signal

$$B_k(s;X,W') = \mathbb{E}_{C_2}\left[L_{C_2,k}\,\varphi_2'(H_2(s,C_2;X,W'))\,w_2'(s,C_2)\right].$$

Because φ_2' is bounded, B_k itself does not introduce any additional r factor (it has one fixed column k of L); the only place where r can enter is when controlling H_2 (and hence $\varphi_2(H_2)$ and $\varphi_2'(H_2)$ through Lipschitzness).

Concretely, in the two-layer roadmap of Section 5:

- Lemma C (Forward Lipschitz) uses a bound of the form

  $$|H_2(W') - H_2(W'')| \leq \|L\|_{\infty,1}\max_k|m_k(W') - m_k(W'')| \leq K\,\|L\|_{\infty,1}\,\|w_1' - w_1''\|_t,$$

  so the constant in Lemma C is multiplied by at most $\|L\|_{\infty,1}$ (hence at most rK under entrywise control).

- Lemma D (Backward Lipschitz) inherits the same multiplicative factor when bounding drift differences involving $\varphi_2(H_2)$ or $\varphi_2'(H_2)$, because those differences are controlled by Lipschitz constants times $|H_2(W') - H_2(W'')|$ on the good event.

Therefore the difference inequality of the form

$$\|F(W') - F(W'')\|_t \leq C\int_0^t \Big((1+B)\,\|W' - W''\|_s + e^{-cB^2}\Big)\,ds$$

remains valid with the same structure, after replacing C by $C\,\mathrm{poly}(\|L\|_{\infty,1})$. The Picard iteration and the $B = \sqrt{m}$ choice are unchanged; only the polynomial function $K_0(t)$ (and hidden constants) now depend on $\|L\|_{\infty,1}$, i.e. on r through $\|L\|_{\infty,1} \leq r\,\sup_{c_2,k}|L_{c_2,k}|$.

2 Channel-wise partial functions via pushforward and disintegration

This section rewrites the r channel summaries $m_k(t;\cdot)$ as integrals against a measure μ_0 on the *untrained* first-layer features, together with a pushforward (or, equivalently, a backward kernel) that incorporates the *trained* first-layer weights w_1.

2.1 Untrained features and their pushforward law

Let $w_0 : \Omega_1 \to \mathbb{R}^d$ denote the untrained first-layer map (in the main note, $w_0 = f_1$) and let ρ^1 be the law of C_1. Define the pushforward measure on feature space

$$\mu_0 \equiv (w_0)_{\#}\rho^1 \qquad \text{on } \mathbb{R}^d,$$

meaning that for any measurable $A \subseteq \mathbb{R}^d$, $\mu_0(A) = \rho^1(\{c_1 : w_0(c_1) \in A\})$.

2.2 The r partial functions as integrals over μ_0

For each channel $k \in \{1, \dots, r\}$, recall the partial function

$$m_k(t; X, W) = \mathbb{E}_{C_1}[w_1(t, C_1, k)\, \varphi_1(\langle w_0(C_1),\, X\rangle)]\,.$$

It is often useful to view $m_k(t;\cdot)$ as an integral over μ_0 by packaging w_1 into a signed (or weighted) pushforward measure.

Definition 1 (Signed pushforward incorporating w_1)**.** *For each $t \geq 0$ and each $k \in \{1, \dots, r\}$, define the (finite signed) measure $\mu^t_{1,k}$ on $\mathbb{R}^d$ by*

$$\mu^t_{1,k} \equiv (w_0)_{\#}\big(w_1(t, \cdot, k)\, \rho^1\big),$$

i.e. for any measurable $A \subseteq \mathbb{R}^d$,

$$\mu^t_{1,k}(A) = \int_{\Omega_1} \mathbb{1}\{w_0(c_1) \in A\}\, w_1(t, c_1, k)\, \rho^1(dc_1).$$

Lemma 3 (Integral representation over feature space)**.** *For every $t \geq 0$, $k \in \{1, \dots, r\}$ and input X,*

$$m_k(t; X, W) = \int_{\mathbb{R}^d} \varphi_1(\langle \theta,\, X\rangle)\, \mu^t_{1,k}(d\theta).$$

2.3 Backward kernel (disintegration) form

If one prefers to keep μ_0 fixed and represent only a *density* on top of it, one can use a disintegration of ρ^1 along the map w_0. There exists a measurable Markov kernel $\kappa(dc_1 \mid \theta)$ on Ω_1 such that

$$\rho^1(dc_1) = \int_{\mathbb{R}^d} \kappa(dc_1 \mid \theta)\, \mu_0(d\theta), \qquad \kappa\big(\{c_1 : w_0(c_1) = \theta\} \mid \theta\big) = 1 \text{ for } \mu_0\text{-a.e. } \theta.$$

Define the channel-wise conditional average of w_1 at feature location θ by

$$\bar{w}_{1,k}(t, \theta) \equiv \int_{\Omega_1} w_1(t, c_1, k)\, \kappa(dc_1 \mid \theta).$$

Then $\mu_{1,k}^t$ is absolutely continuous with respect to μ_0 with Radon–Nikodym derivative $\bar{w}_{1,k}(t,\cdot)$, in the sense that

$$\mu_{1,k}^t(d\theta) = \bar{w}_{1,k}(t,\theta)\,\mu_0(d\theta),$$

and hence the kth partial function can be written as the μ_0-integral

$$m_k(t;X,W) = \int_{\mathbb{R}^d} \bar{w}_{1,k}(t,\theta)\,\varphi_1(\langle\theta,\,X\rangle)\,\mu_0(d\theta).$$

Remark 2 (Interpretation and special cases)**.** *The family $(m_k(t;\cdot))_{k=1}^r$ can be seen as r scalar functions generated by the* same *untrained feature measure μ_0 and r channel-dependent coefficient fields $\bar{w}_{1,k}(t,\cdot)$ on feature space. If w_0 is injective, then $\kappa(dc_1 \mid \theta)$ is a Dirac mass at $c_1 = w_0^{-1}(\theta)$ and $\bar{w}_{1,k}(t,\theta) = w_1(t, w_0^{-1}(\theta), k)$.*

3 Time variation of the r partial functions and the induced PDE

Define for each channel $k \in \{1,\dots,r\}$ the partial function on input space

$$f_k(t,x) \equiv m_k(t;x,W) = \int_{\mathbb{R}^d} \bar{w}_{1,k}(t,\theta)\,\varphi_1(\langle\theta,\,x\rangle)\,\mu_0(d\theta).$$

We compute its time derivative and state the (measure-valued) evolution equation implied by the MF ODE for w_1.

3.1 Evolution of the coefficient field $\bar{w}_{1,k}(t,\theta)$

From the MF ODE (cf. (**??**) in the main note), the first-layer weights satisfy

$$\partial_t w_1(t,c_1,k) = -\,\xi_1(t)\,\mathbb{E}_Z\Big[d_L\big(Z;W(t)\big)\,\varphi_1(\langle w_0(c_1),\,X\rangle)\,\mathbb{E}_{C_2}\Big[L_{C_2,k}\,\varphi_2'\Big(H_2(t,C_2;X,W(t))\Big)\,w_2(t,C_2)\Big]\Big].$$

Introduce the channel-k backpropagated signal (a random variable through X)

$$B_k(t;X) \equiv \mathbb{E}_{C_2}\Big[L_{C_2,k}\,\varphi_2'\Big(H_2(t,C_2;X,W(t))\Big)\,w_2(t,C_2)\Big].$$

Since the drift depends on c_1 only through $\theta = w_0(c_1)$, the conditional average $\bar{w}_{1,k}(t,\theta)$ evolves pointwise as

$$\partial_t \bar{w}_{1,k}(t,\theta) = -\xi_1(t)\,\mathbb{E}_{Z=(X,Y)}\big[d_L\big(Z;W(t)\big)\,\varphi_1(\langle\theta,\,X\rangle)\,B_k(t;X)\big]. \tag{1}$$

3.2 A measure-valued PDE for $\mu_{1,k}^t$

Recall $\mu_{1,k}^t(d\theta) = \bar{w}_{1,k}(t,\theta)\,\mu_0(d\theta)$. Equation (1) is equivalently the following linear "reaction" PDE for the signed measure $\mu_{1,k}^t$:

$$\partial_t \mu_{1,k}^t(d\theta) = -\xi_1(t)\,\mathbb{E}_{Z=(X,Y)}\big[d_L\big(Z;W(t)\big)\,\varphi_1(\langle\theta,\,X\rangle)\,B_k(t;X)\big]\,\mu_0(d\theta). \tag{2}$$

In weak form, for every bounded measurable test function $\psi : \mathbb{R}^d \to \mathbb{R}$,

$$\frac{d}{dt}\int_{\mathbb{R}^d} \psi(\theta)\,\mu_{1,k}^t(d\theta) = -\xi_1(t)\,\mathbb{E}_{Z=(X,Y)}\left[d_L\big(Z;W(t)\big)\,B_k(t;X)\int_{\mathbb{R}^d}\psi(\theta)\,\varphi_1(\langle\theta,\,X\rangle)\,\mu_0(d\theta)\right].$$

Because w_0 is frozen, there is no transport term (no $\nabla_\theta \cdot (\cdot)$): the feature locations stay fixed and only the signed density evolves.

3.3 Variation of $f_k(t,x)$ and an induced integro-differential equation

Differentiating under the integral and using (1) yields

$$\begin{aligned}\partial_t f_k(t,x) &= \int_{\mathbb{R}^d} \partial_t \bar{w}_{1,k}(t,\theta)\, \varphi_1(\langle\theta,\, x\rangle)\, \mu_0(d\theta) \\ &= -\xi_1(t)\, \mathbb{E}_{Z=(X,Y)}\Big[d_L\big(Z;W(t)\big)\, B_k(t;X) \int_{\mathbb{R}^d} \varphi_1(\langle\theta,\, X\rangle)\, \varphi_1(\langle\theta,\, x\rangle)\, \mu_0(d\theta)\Big]. \end{aligned} \tag{3}$$

Define the kernel induced by the untrained feature measure μ_0:

$$K_{\mu_0}(x,x') \equiv \int_{\mathbb{R}^d} \varphi_1(\langle\theta,\, x\rangle)\, \varphi_1(\langle\theta,\, x'\rangle)\, \mu_0(d\theta).$$

Then (3) becomes the compact evolution equation

$$\partial_t f_k(t,x) = -\xi_1(t)\, \mathbb{E}_{Z=(X,Y)}\big[d_L\big(Z;W(t)\big)\, B_k(t;X)\, K_{\mu_0}(x,X)\big], \qquad k=1,\dots,r. \tag{4}$$

This is an r-coupled integro-differential system in t (and in x through the kernel), closed once one expresses $B_k(t;X)$ and $d_L(Z;W(t))$ in terms of the network output $\hat{y}(\cdot;W(t))$, which itself depends on $(f_j)_{j=1}^r$ through $H_2 = \sum_{j=1}^r L_{\cdot,j} f_j$.

3.4 ReLU top layer: sign-coherent gating and a reinforcement heuristic

Assume the second-layer activation is ReLU, $\varphi_2(u) = u_+$, so that $\varphi_2'(u) = \mathbb{1}\{u>0\}$ a.e. Then the channel-k backpropagated signal becomes

$$B_k(t;X) = \mathbb{E}_{C_2}[L_{C_2,k}\, w_2(t,C_2)\, \mathbb{1}\{H_2(t,C_2;X,W(t))>0\}]. \tag{5}$$

In words: $B_k(t;X)$ is a *gated mixture* of the kth column of L, weighted by the current top-layer weights w_2, with the gate determined by H_2.

Combining (4) with (5) shows that, for each k,

$$\partial_t f_k(t,x) = -\xi_1(t)\, \mathbb{E}_{Z=(X,Y)}\big[d_L\big(Z;W(t)\big)\, K_{\mu_0}(x,X)\, \mathbb{E}_{C_2}[L_{C_2,k}\, w_2(t,C_2)\, \mathbb{1}\{H_2(t,C_2;X,W(t))>0\}]\big].$$

Remark 3 (When does "large f_j makes itself larger" hold?)**.** *The intuition that a comparatively large f_j amplifies its own future growth goes through a* sign-coherence *regime:*

- *if $L_{c_2,k} \geq 0$ (or at least does not change sign too much) for the relevant c_2,*
- *if $w_2(t,c_2) \geq 0$ (or has a controlled sign pattern aligned with $L_{c_2,\cdot}$),*
- *if the loss derivative $d_L(Z;W(t))$ has a consistent sign on the region of inputs currently being learned (e.g. in a residual-fitting phase for square loss),*

then increasing one partial function $f_j(t,\cdot)$ tends to increase H_2 on the subset of (C_2,X) where $L_{C_2,j} > 0$, which increases the gate $\mathbb{1}\{H_2>0\}$ and hence increases $B_j(t;X)$. Since $\partial_t f_j$ is proportional to $B_j(t;X)$ through (4), this creates a positive feedback loop in that region.

Without such sign restrictions, reinforcement is not guaranteed: w_2 and d_L can change sign, and $L_{c_2,k}$ may have mixed signs, so the same mechanism can produce cancellation or even competition between channels.

3.5 Fixed random-sign L: an "emergent" sign-coherence in a one-spike reduction

The sign-coherence bullets in Remark 3 are sufficient, but in practice $L_{c_2,k}$ can have mixed signs (e.g. random at initialization and frozen). In that case, one can still obtain an *effective* sign-coherence once the dynamics at a given input location is dominated by a single channel. We explain the mechanism in a simplified "single spike" reduction.

Single spike reduction. Assume the input distribution is concentrated at one point $x_\star$ (or that we are in a local-kernel regime where the dynamics at $x_\star$ decouple), and set

$$f_k(t) \equiv f_k(t, x_\star), \qquad d(t) \equiv d_L\big((x_\star, y_\star); W(t)\big), \qquad L_{c_2} \equiv (L_{c_2,1}, \dots, L_{c_2,r}) \in \mathbb{R}^r.$$

With ReLU $\varphi_2(u) = u_+$, the gate is $\mathbb{1}\{\langle L_{c_2},\, f(t)\rangle > 0\}$ and the two relevant MF equations become

$$\partial_t f_k(t) = -\xi_1(t)\, d(t)\, B_k(t), \tag{6}$$
$$B_k(t) = \mathbb{E}_{C_2}[L_{C_2,k}\, w_2(t, C_2)\, \mathbb{1}\{\langle L_{C_2},\, f(t)\rangle > 0\}]\,, \qquad k = 1, \dots, r, \tag{7}$$
$$\partial_t w_2(t, c_2) = -\xi_2(t)\, d(t)\, \big(\langle L_{c_2},\, f(t)\rangle\big)_+. \tag{8}$$

If $w_2(0, \cdot) \geq 0$ and $-d(t) \geq 0$ (for instance in an underfitting phase for square loss), then (8) implies $w_2(t, \cdot)$ remains nonnegative and increases more for those c_2 with larger positive $\langle L_{c_2},\, f(t)\rangle$.

Why mixed signs in L may still reinforce a dominant channel. Suppose at some time t the vector $f(t)$ is (approximately) one-sparse,

$$f(t) \approx a(t)\, e_j, \qquad a(t) \in \mathbb{R}, \quad j \in \{1, \dots, r\}.$$

Then $\langle L_{c_2},\, f(t)\rangle \approx a(t)\, L_{c_2,j}$ and the gate becomes $\mathbb{1}\{a(t) L_{c_2,j} > 0\}$. If (heuristically, or in an early-time approximation) $w_2(t, C_2)$ is not strongly *negatively* correlated with the sign of $L_{C_2,j}$, then conditioning on the gate makes the jth coordinate have a *nonzero mean*:

$$\mathbb{E}[L_{C_2,j}\, \mathbb{1}\{a L_{C_2,j} > 0\}] = \operatorname{sign}(a)\, \mathbb{E}[|L_{C_2,j}|]\, /2,$$

whereas for $k \neq j$, under approximate independence and symmetry across coordinates of L_{C_2},

$$\mathbb{E}[L_{C_2,k}\, \mathbb{1}\{a L_{C_2,j} > 0\}] \approx 0.$$

Plugging this into (7) suggests that, even when L has mixed signs, the gate selects a half-space primarily determined by the dominant coordinate j, and this selection biases $B_j(t)$ to have the *same sign* as $a(t)$ while keeping $B_k(t)$ small for $k \neq j$. Equation (6) then amplifies the dominant channel at that spike location. Moreover, (8) further increases $w_2(t, c_2)$ on the gated set $\{c_2 : \langle L_{c_2},\, f(t)\rangle > 0\}$, strengthening the conditional bias in (7).

A rigorous one-sparse statement. The preceding intuition can be made fully rigorous in the exact one-sparse regime, under a mild symmetry/independence assumption on the frozen random vector L_{C_2} and a sign-consistent residual.

Assumption 1 (Random mixing vector, symmetric and independent). *The random vector $L_{C_2} = (L_{C_2,1}, \dots, L_{C_2,r})$ has independent coordinates, each symmetric about 0: $L_{C_2,k} \overset{d}{=} -L_{C_2,k}$ and $\mathbb{E}[L_{C_2,k}^2] < \infty$ for all k.*

Lemma 4 (Half-space symmetry identities). *Let U be a real random variable with $U \stackrel{d}{=} -U$ and $\mathbb{E}[|U|] < \infty$. Then for any $a \in \mathbb{R} \setminus \{0\}$,*

$$\mathbb{E}[U\,\mathbb{1}\{aU > 0\}] = \operatorname{sign}(a)\,\frac{1}{2}\mathbb{E}[|U|].$$

If moreover $\mathbb{E}[U^2] < \infty$, then

$$\mathbb{E}\big[U^2\,\mathbb{1}\{aU > 0\}\big] = \frac{1}{2}\mathbb{E}[U^2].$$

Proof. By symmetry $U \stackrel{d}{=} -U$, we have $\mathbb{E}[U] = 0$ (when $\mathbb{E}[|U|] < \infty$). For $a > 0$, decompose

$$0 = \mathbb{E}[U] = \mathbb{E}[U\mathbb{1}\{U > 0\}] + \mathbb{E}[U\mathbb{1}\{U < 0\}],$$

so $\mathbb{E}[U\mathbb{1}\{U > 0\}] = -\mathbb{E}[U\mathbb{1}\{U < 0\}]$. Moreover,

$$\mathbb{E}[|U|] = \mathbb{E}[U\mathbb{1}\{U > 0\}] - \mathbb{E}[U\mathbb{1}\{U < 0\}] = 2\,\mathbb{E}[U\mathbb{1}\{U > 0\}],$$

which gives $\mathbb{E}[U\mathbb{1}\{U > 0\}] = \mathbb{E}[|U|]/2$. For $a < 0$, $\mathbb{1}\{aU > 0\} = \mathbb{1}\{U < 0\}$ and the identity becomes $\mathbb{E}[U\mathbb{1}\{U < 0\}] = -\mathbb{E}[|U|]/2$. This yields $\mathbb{E}[U\mathbb{1}\{aU > 0\}] = \operatorname{sign}(a)\mathbb{E}[|U|]/2$ for all $a \neq 0$. The second-moment identity follows similarly from symmetry: $U^2\mathbb{1}\{U > 0\}$ and $U^2\mathbb{1}\{U < 0\}$ have equal expectation, so $\mathbb{E}[U^2\mathbb{1}\{U > 0\}] = \mathbb{E}[U^2]/2$ and likewise for $\mathbb{1}\{aU > 0\}$. □

Lemma 5 (One-sparse invariant manifold and emergent sign-coherence). *Assume the one-spike system (6)–(8), Assumption 1, and that the residual has fixed sign on $[0, T]$: $d(t) \leq 0$ for all $t \in [0, T]$. Assume a nonnegative constant initialization in the top layer, $w_2(0, c_2) \equiv w_2^0 \geq 0$. Fix $j \in \{1, \ldots, r\}$ and assume a one-sparse spike at $t = 0$ with positive amplitude, $f(0) = a_0 e_j$ with $a_0 > 0$. Then for all $t \in [0, T]$:*

- *$f(t) = a(t)e_j$ for some scalar $a(t) \geq a_0$;*
- *for $k \neq j$, $B_k(t) = 0$;*
- *$B_j(t) > 0$ and admits the explicit representation*

$$B_j(t) = \frac{w_2^0}{2}\mathbb{E}[|L_{C_2,j}|] \; + \; \frac{\mathbb{E}[L_{C_2,j}^2]}{2}\int_0^t \xi_2(s)\,(-d(s))\,a(s)\,ds. \tag{9}$$

Consequently $\partial_t a(t) = -\xi_1(t)\,d(t)\,B_j(t) \geq 0$ on $[0, T]$.

Proof. We work on $[0, T]$ and write $U \equiv L_{C_2,j}$. Assume inductively that $f(t) = a(t)e_j$ with $a(t) \geq 0$ (this will be justified by the conclusion). Then $\langle L_{c_2}, f(t)\rangle = a(t)L_{c_2,j}$ and the gate $\mathbb{1}\{\langle L_{C_2}, f(t)\rangle > 0\}$ is $\mathbb{1}\{U > 0\}$. The top-layer ODE (8) integrates explicitly to

$$w_2(t, c_2) = w_2^0 + \Big(\int_0^t \xi_2(s)\,(-d(s))\,a(s)\,ds\Big)\,L_{c_2,j}\,\mathbb{1}\{L_{c_2,j} > 0\},$$

since $(a(s)L_{c_2,j})_+ = a(s)L_{c_2,j}\mathbb{1}\{L_{c_2,j} > 0\}$ for $a(s) \geq 0$. In particular $w_2(t, C_2)$ is a measurable function of $L_{C_2,j}$ only.

For $k \neq j$, independence of coordinates implies $L_{C_2,k}$ is independent of $(L_{C_2,j}, w_2(t, C_2), \mathbb{1}\{U > 0\})$, hence

$$B_k(t) = \mathbb{E}[L_{C_2,k}\,w_2(t, C_2)\,\mathbb{1}\{U > 0\}] = \mathbb{E}[L_{C_2,k}]\,\mathbb{E}[w_2(t, C_2)\,\mathbb{1}\{U > 0\}] = 0,$$

using $\mathbb{E}[L_{C_2,k}] = 0$ from symmetry. Thus $\partial_t f_k(t) = -\xi_1(t)d(t)B_k(t) = 0$ for all $k \neq j$, so $f_k(t) \equiv 0$ on $[0,T]$. This proves invariance of the one-sparse manifold.

For $k = j$, let $\alpha(t) \equiv \int_0^t \xi_2(s)\,(-d(s))\,a(s)\,ds$. Then the above formula gives $w_2(t, C_2) = w_2^0 + \alpha(t)\,U\,\mathbb{1}\{U > 0\}$ and

$$B_j(t) = \mathbb{E}\big[U\,(w_2^0 + \alpha(t)U\mathbb{1}\{U > 0\})\,\mathbb{1}\{U > 0\}\big] = w_2^0\,\mathbb{E}[U\mathbb{1}\{U > 0\}] + \alpha(t)\,\mathbb{E}[U^2\mathbb{1}\{U > 0\}].$$

Applying Lemma 4 with $a = 1$ yields (9) and in particular $B_j(t) > 0$. Finally, since $f_j(t) = a(t)$ and $\partial_t f_j(t) = -\xi_1(t)d(t)B_j(t)$, we have $\partial_t a(t) \geq 0$ because $d(t) \leq 0$ and $B_j(t) > 0$. Thus $a(t) \geq a_0 > 0$ for all $t \in [0,T]$, closing the argument. $\square$

Remark 4 (Interpretation of Lemma 5)**.** *Even if $L_{C_2,j}$ has mixed signs, once $f(t) = a(t)e_j$ with $a(t) > 0$ the gate is exactly $\mathbb{1}\{L_{C_2,j} > 0\}$ and $w_2(t, c_2)$ increases on that half-space via* (8)*. Lemma 4 then implies a strictly positive bias in $B_j(t)$, while for $k \neq j$ the independence and symmetry enforce $B_k(t) = 0$.*

Takeaway. In this one-spike regime, *random sign patterns do not necessarily destroy reinforcement*; instead, once one channel dominates, the ReLU gate can create an emergent sign-coherence by conditioning on a half-space determined by that dominant coordinate. This provides a plausible route to winner-take-all specialization without requiring $L_{c_2,k} \geq 0$ for all k.

Remark 5 (Heuristic link to specialization and "spikes")**.** *In one-dimensional problems, the gate $\mathbb{1}\{H_2(t, C_2; x) > 0\}$ can partition the input axis into regions that activate different subsets of c_2 (hence different linear combinations of the r channels through L). If different channels k become dominant on different regions,* (4) *shows they are trained using different effective signed measures over X, yielding a self-reinforcing* specialization *in distinct "slots". This is a plausible mechanism for learning localized high-amplitude structures (spikes) at different locations, driven by the gating nonlinearity rather than by a globally smooth kernel alone.*

4 Toward a proof of spike specialization: a workable program

This section records a proof "démarche" for the empirically observed phenomenon: different channels $k \in \{1, \ldots, r\}$ learn localized high-amplitude structures at different input locations ("spike specialization"). The goal is to isolate statements that are both meaningful and provable from the dynamics (4)–(5).

4.1 Step 0: formalize what "specialization" and "spike" mean

There are at least two mathematically tractable targets.

(A) Channel specialization (winner-take-all at each location). Define the channel shares

$$s_k(t, x) \equiv \frac{|f_k(t, x)|}{\sum_{j=1}^r |f_j(t, x)| + \varepsilon}, \qquad k = 1, \ldots, r,$$

with $\varepsilon > 0$ for stability. One aims to show that for many x there exists $k^\star(x)$ such that $s_{k^\star(x)}(t, x) \to 1$ and $s_k(t, x) \to 0$ for $k \neq k^\star(x)$. This captures specialization without committing to a particular notion of spike.

(B) Spike formation (localization / high-frequency growth). One aims to show that for some channels k, the function $f_k(t,\cdot)$ becomes localized, e.g. $\|f_k(t,\cdot)\|_\infty$ grows while $\|f_k(t,\cdot)\|_{L^1}$ stays controlled, or that a suitable high-frequency energy increases. This typically requires additional structure on μ_0 and φ_1 (e.g. wavelet-like features).

4.2 Step 1: reduce to an empirical (finite-sample) dynamical system

Let $x_1, \dots, x_n$ be the training inputs and consider the finite vector process

$$F_k(t) \equiv \big(f_k(t,x_1), \dots, f_k(t,x_n)\big) \in \mathbb{R}^n, \qquad k = 1, \dots, r.$$

If one replaces $\mathbb{E}_X[\cdot]$ by the empirical average $n^{-1}\sum_{i=1}^n(\cdot)\big|_{X=x_i}$, then (4) yields an ODE on $\mathbb{R}^{r\times n}$:

$$\partial_t f_k(t,x_i) = -\xi_1(t)\,\frac{1}{n}\sum_{j=1}^n d_L\big((x_j,y_j);W(t)\big)\,B_k(t;x_j)\,K_{\mu_0}(x_i,x_j).$$

This finite-dimensional setting is a natural starting point for a rigorous analysis of specialization, and it matches what is observed experimentally.

4.3 Step 2: identify the "local" regime (nearly diagonal kernel)

The kernel

$$K_{\mu_0}(x,x') = \int_{\mathbb{R}^d} \varphi_1(\langle\theta,\,x\rangle)\,\varphi_1(\langle\theta,\,x'\rangle)\,\mu_0(d\theta)$$

controls how strongly the update at x depends on data at X. If K_{μ_0} is localized (as in many wavelet constructions), then in a suitable grid setting one can be close to a diagonal-dominant regime,

$$K_{\mu_0}(x_i,x_j) \approx 0 \quad \text{for } i \neq j,$$

so each location x_i is driven primarily by its own residual and gating. In the ideal limit $K_{\mu_0}(x_i,x_j) = \delta_{ij}$, the dynamics decouple across i, and specialization can be studied pointwise in x. The general case can then be treated as a perturbation of this local regime.

4.4 Step 3: make reinforcement into a sign-coherent inequality

The essential nonlinear feedback comes from the ReLU gate inside B_k:

$$B_k(t;X) = \mathbb{E}_{C_2}[L_{C_2,k}\,w_2(t,C_2)\,\mathbb{1}\{H_2(t,C_2;X) > 0\}], \qquad H_2(t,c_2;X) = \sum_{j=1}^r L_{c_2,j}\,f_j(t,X).$$

To prove a monotone reinforcement effect ("a larger f_j tends to increase its own future growth"), one typically restricts to a sign-coherent regime where:

- $L_{c_2,k} \geq 0$ on the relevant c_2-mass (or has controlled sign changes),
- $w_2(t,c_2) \geq 0$ (or is aligned with $L_{c_2,\cdot}$),
- the residual signal $d_L(Z;W(t))$ has a stable sign on the region of X currently being fit (for instance locally in time or under one-sided targets).

Under such conditions, increasing $f_j(t,X)$ increases $H_2(t,C_2;X)$ on the set where $L_{C_2,j} > 0$, hence (weakly) increases the gate $\mathbb{1}\{H_2 > 0\}$, hence increases $B_j(t;X)$, closing a positive feedback loop through (4).

4.5 Step 4: prove specialization via relative-growth (replicator-style) dynamics

In a local regime (Step 2), define at fixed x the log-ratio

$$R_{k\ell}(t,x) \equiv \log \frac{|f_k(t,x)| + \varepsilon}{|f_\ell(t,x)| + \varepsilon}.$$

The objective is to show that for most (k, ℓ, x) one has a differential inequality of the form

$$\partial_t R_{k\ell}(t,x) \geq \mathrm{Adv}_{k\ell}(t,x) - \mathrm{Err}(t,x),$$

where the "advantage" term $\mathrm{Adv}_{k\ell}$ is increasing in the current dominance of channel k at x due to the gating in B_k, and Err is controlled by off-diagonal kernel interactions and bad-event tails. Such inequalities yield exponential separation of ratios and thus winner-take-all channel shares $s_k(t,x)$.

4.6 Step 4.5: log-ratio calculus (exact identity) and a sufficient condition

Fix x and define $u_k(t,x) \equiv |f_k(t,x)| + \varepsilon$. Whenever $f_k(\cdot, x)$ is absolutely continuous in t, the map $u_k(\cdot, x)$ is also absolutely continuous and, for almost every t,

$$\partial_t u_k(t,x) = \mathrm{sign}(f_k(t,x))\, \partial_t f_k(t,x) \qquad \text{whenever } f_k(t,x) \neq 0,$$

and any choice in the interval $[-|\partial_t f_k(t,x)|, |\partial_t f_k(t,x)|]$ is a valid subderivative when $f_k(t,x) = 0$. In particular, for almost every t one has the exact log-ratio identity

$$\partial_t R_{k\ell}(t,x) = \frac{\mathrm{sign}(f_k(t,x))\, \partial_t f_k(t,x)}{|f_k(t,x)| + \varepsilon} - \frac{\mathrm{sign}(f_\ell(t,x))\, \partial_t f_\ell(t,x)}{|f_\ell(t,x)| + \varepsilon}. \tag{10}$$

Local decoupling and per-capita gains. In the ideal local regime $K_{\mu_0}(x, X) \approx K_{\mu_0}(x,x)\, \mathbb{1}\{X = x\}$ (or in the diagonal empirical-kernel approximation at a training point $x = x_i$), (4) reduces to

$$\partial_t f_k(t,x) \approx -\xi_1(t)\, K_{\mu_0}(x,x) \Big(d_x(t)\, B_k(t;x) \Big), \tag{11}$$

where $d_x(t)$ abbreviates the local residual signal (e.g. $d_L((x,y); W(t))$ in the one-point reduction). Plugging (11) into (10) motivates the *per-capita gain*

$$\gamma_k(t,x) \equiv \xi_1(t)\, K_{\mu_0}(x,x)\, \frac{-d_x(t)\, \mathrm{sign}(f_k(t,x))\, B_k(t;x)}{|f_k(t,x)| + \varepsilon},$$

so that $\partial_t R_{k\ell}(t,x) \approx \gamma_k(t,x) - \gamma_\ell(t,x)$. Thus, specialization at x is implied by a persistent gap in per-capita gains.

Lemma 6 (A sufficient per-capita gap condition)**.** *Fix x and suppose there exist $\delta > 0$ and an index j such that on a time interval $[t_0, t_1]$,*

$$\gamma_j(t,x) \geq \gamma_k(t,x) + \delta \qquad \textit{for all } k \neq j \textit{ and almost every } t \in [t_0, t_1].$$

Then for every $k \neq j$,

$$R_{jk}(t_1, x) \geq R_{jk}(t_0, x) + \delta\, (t_1 - t_0),$$

and hence the share of channel j at x increases relative to all other channels over $[t_0, t_1]$.

Example: the one-sparse spike regime. In the one-spike reduction (6)–(8) under the hypotheses of Lemma 5, we have $f(t) = a(t)e_j$ with $a(t)$ nondecreasing and $B_k(t) = 0$ for $k \neq j$. Then for $k \neq j$ the denominator $|f_k| + \varepsilon = \varepsilon$ is constant while $|f_j| + \varepsilon = a(t) + \varepsilon$ increases, so the log-ratio satisfies

$$\partial_t R_{jk}(t) = \frac{\partial_t a(t)}{a(t) + \varepsilon} = \frac{-\xi_1(t)\, d(t)\, B_j(t)}{a(t) + \varepsilon} > 0,$$

illustrating concretely how reinforcement yields monotone growth of log-ratios in a spike-dominated regime.

Remark 6 (The "minus sign" in $\partial_t f_k = -\xi_1 d\, B_k$ does not break specialization)**.** *The minus sign is simply the gradient-descent direction. It does* not *turn reinforcement into anti-reinforcement by itself; what matters is the sign of the residual signal $d_x(t)$. For instance, for square loss $\mathcal{L}(y, \hat{y}) = \frac{1}{2}(\hat{y} - y)^2$, one has $d_x(t) = \hat{y}(x; W(t)) - y(x)$, which typically changes sign when the prediction crosses the target. Thus the spike amplitude $|f_j(t, x)|$ need not increase monotonically for all times. The robust specialization statement is instead about* relative dominance*: log-ratios $R_{jk}(t, x)$ can grow even during phases where all $|f_k(t, x)|$ decrease, because R_{jk} tracks* per-capita *gains (Step 4.5) rather than absolute growth.*

4.7 Step 4.6: effective sharpness along the spike direction (one-point square loss)

In a one-point reduction, one can make the intuition "the descent direction is clear" fully quantitative via a Lyapunov identity. Assume a single training pair $(x_\star, y_\star)$ with square loss $\mathcal{L} = \frac{1}{2}(\hat{y} - y_\star)^2$, and that we are on the one-sparse manifold of Lemma 5 so that $f(t) = a(t)e_j$ with $a(t) \geq 0$ and $\hat{y}(t) = a(t)B_j(t)$. Define the residual $d(t) \equiv \hat{y}(t) - y_\star$ and the scalar loss $\ell(t) \equiv \frac{1}{2}d(t)^2$.

Lemma 7 (Lyapunov identity and an "effective curvature" coefficient)**.** *Assume the one-spike system* (6)–(8)*, Assumption 1, and the one-sparse manifold $f(t) = a(t)e_j$ with $a(t) \geq 0$ on a time interval where the gate is $\mathbb{1}\{L_{C_2,j} > 0\}$. For square loss, the residual satisfies the closed scalar differential equation*

$$\partial_t d(t) = -d(t)\Big(\xi_1(t)\, B_j(t)^2 + \xi_2(t)\, a(t)^2\, S_j\Big), \qquad S_j \equiv \mathbb{E}\big[L_{C_2,j}^2\, \mathbb{1}\{L_{C_2,j} > 0\}\big]\,. \tag{12}$$

Consequently the loss decreases according to

$$\partial_t \ell(t) = -2\, \ell(t)\Big(\xi_1(t)\, B_j(t)^2 + \xi_2(t)\, a(t)^2\, S_j\Big) \leq 0. \tag{13}$$

Proof. Under the one-sparse manifold $f(t) = a(t)e_j$ with $a(t) \geq 0$, the network prediction at $x_\star$ is

$$\hat{y}(t) = \mathbb{E}_{C_2}[w_2(t, C_2)\, (a(t)L_{C_2,j})_+] = a(t)\, \mathbb{E}_{C_2}[w_2(t, C_2)\, L_{C_2,j}\, \mathbb{1}\{L_{C_2,j} > 0\}] = a(t)\, B_j(t),$$

so $d(t) = a(t)B_j(t) - y_\star$. Differentiate:

$$\partial_t d(t) = \partial_t (a(t)B_j(t)) = (\partial_t a(t))B_j(t) + a(t)\, \partial_t B_j(t).$$

By (6), $\partial_t a(t) = \partial_t f_j(t) = -\xi_1(t)\, d(t)\, B_j(t)$. Moreover, by definition $B_j(t) = \mathbb{E}[L_{C_2,j}\, w_2(t, C_2)\, \mathbb{1}\{L_{C_2,j} > 0\}]$ on this manifold, hence

$$\partial_t B_j(t) = \mathbb{E}[L_{C_2,j}\, (\partial_t w_2(t, C_2))\, \mathbb{1}\{L_{C_2,j} > 0\}]\,.$$

Using (8) with $\langle L_{c_2},\, f(t)\rangle = a(t)L_{c_2,j}$ and $a(t) \geq 0$ gives $(\langle L_{C_2},\, f(t)\rangle)_+ = a(t)L_{C_2,j}\mathbb{1}\{L_{C_2,j} > 0\}$, so

$$\partial_t w_2(t, C_2) = -\xi_2(t)\, d(t)\, a(t)\, L_{C_2,j}\, \mathbb{1}\{L_{C_2,j} > 0\}.$$

Therefore

$$\partial_t B_j(t) = -\xi_2(t)\, d(t)\, a(t)\, \mathbb{E}\big[L_{C_2,j}^2\, \mathbb{1}\{L_{C_2,j} > 0\}\big] = -\xi_2(t)\, d(t)\, a(t)\, S_j.$$

Substituting the expressions for $\partial_t a$ and $\partial_t B_j$ yields

$$\partial_t d(t) = (-\xi_1(t)d(t)B_j(t))B_j(t) + a(t)(-\xi_2(t)d(t)a(t)S_j) = -d(t)\Big(\xi_1(t)B_j(t)^2 + \xi_2(t)a(t)^2 S_j\Big),$$

which is (12). Finally, $\ell(t) = \frac{1}{2}d(t)^2$ implies $\partial_t \ell(t) = d(t)\partial_t d(t)$, and (13) follows. □

Remark 7 (What "sharpness" means here)**.** *The coefficient $\lambda(t) \equiv \xi_1(t)B_j(t)^2 + \xi_2(t)a(t)^2 S_j$ plays the role of an* effective curvature along the active spike direction*. If $\lambda(t) \geq \lambda_0 > 0$ on an interval, then* (13) *gives exponential decay $\ell(t) \leq \ell(t_0)e^{-2\lambda_0(t-t_0)}$. This is a rigorous version of "the landscape is sharp in the useful direction". It should not be confused with global strong convexity: low-rank / bilinear parametrizations can retain flat directions (e.g. scaling symmetries) even while being very sharp along the spike-aligned direction.*

4.8 Step 5: connect specialization to spikes using the feature dictionary

Specialization alone does not force spikes; the spike shape is controlled by (μ_0, φ_1) through K_{μ_0}. For wavelet-like features (or other localized dictionaries), channel specialization on disjoint regions implies that different channels are effectively solving different local regression problems, which can produce localized high-amplitude fits (spikes) at different locations. Thus a practical strategy is:

- prove channel specialization (Step 4) in a sign-coherent, local-kernel regime,
- then quantify how localization properties of K_{μ_0} convert specialization into spike-like structure.

4.9 Step 5.1: closed-form dynamics in the kernel (NTK) regime and frequency-by-frequency learning

This subsection records an explicit "solved" dynamics in the regime where the learning dynamics reduce to a *linear* kernel gradient flow with a *fixed* Gram operator. This is the cleanest mathematical lens for understanding learning of localized targets (two-sided steps) or pure frequencies (cosines).

Empirical (finite-sample) kernel dynamics. Fix training points $x_1, \ldots, x_n$ with targets $y = (y_1, \ldots, y_n)^\top$ and let $\hat{y}(t) \in \mathbb{R}^n$ be the vector of current predictions $\hat{y}_i(t) \equiv \hat{y}(x_i; W(t))$. In a standard NTK/random-features linearization (kernel frozen at initialization), the squared-loss gradient flow closes as

$$\partial_t \hat{y}(t) = -\eta(t)\, K\, (\hat{y}(t) - y), \tag{14}$$

where $K \in \mathbb{R}^{n\times n}$ is a fixed positive semidefinite Gram matrix (typically $K_{ij} = K_{\mu_0}(x_i, x_j)$ up to a scaling depending on parametrization) and $\eta(t) \geq 0$ is an effective learning-rate factor.

Lemma 8 (Matrix exponential solution)**.** *Assume K does not depend on t and η is integrable on $[0, \infty)$. Define the effective time $\tau(t) \equiv \int_0^t \eta(s)\, ds$. Then the solution of* (14) *is*

$$\hat{y}(t) - y = \exp\big(-\tau(t)\, K\big)\, (\hat{y}(0) - y). \tag{15}$$

Equivalently, if $K = U\,\mathrm{diag}(\lambda_1, \dots, \lambda_n)\,U^\top$ with $\lambda_i \ge 0$, then the mode coefficients $\alpha_i(t) \equiv u_i^\top(\hat{y}(t) - y)$ satisfy

$$\alpha_i(t) = e^{-\lambda_i\,\tau(t)}\,\alpha_i(0), \qquad i = 1, \dots, n. \tag{16}$$

Proof. Let $e(t) \equiv \hat{y}(t) - y$. Then (14) is $\partial_t e(t) = -\eta(t)Ke(t)$, hence $\partial_\tau e(\tau) = -Ke(\tau)$ in the time-change variable τ. The unique solution is $e(\tau) = e^{-\tau K}e(0)$, which is (15). Diagonalizing K yields (16). □

Continuous kernel flow and Fourier modes (translation-invariant case). To connect directly to "frequency learning", suppose $x \in \mathbb{R}$ (or $x \in \mathbb{T}$) and the data distribution is uniform so that the kernel operator is approximately translation-invariant: $K_{\mu_0}(x, x') = \kappa(x - x')$ for a function κ. Then the squared-loss kernel gradient flow takes the form

$$\partial_t \hat{y}(t, x) = -\eta(t) \int \kappa(x - x')\,(\hat{y}(t, x') - y(x'))\,dx'. \tag{17}$$

Taking Fourier transforms gives a fully decoupled dynamics: for each frequency ω,

$$\partial_t \widehat{(\hat{y} - y)}(t, \omega) = -\eta(t)\,\widehat{\kappa}(\omega)\,\widehat{(\hat{y} - y)}(t, \omega), \qquad \widehat{(\hat{y} - y)}(t, \omega) = e^{-\widehat{\kappa}(\omega)\tau(t)}\,\widehat{(\hat{y} - y)}(0, \omega). \tag{18}$$

Thus each frequency decays exponentially at a rate set by the kernel spectrum $\widehat{\kappa}(\omega)$.

Cosine target (single frequency). If $y(x) = A\cos(\omega_0 x)$ and $\hat{y}(0, \cdot)$ has no ω_0 component, then (18) implies

$$\hat{y}(t, x) = A\big(1 - e^{-\widehat{\kappa}(\omega_0)\tau(t)}\big)\cos(\omega_0 x),$$

so the learning time-scale of frequency ω_0 is exactly $[\widehat{\kappa}(\omega_0)]^{-1}$ (up to the effective time τ). Frequency bias is therefore determined by monotonicity of $\widehat{\kappa}(\omega)$ in $|\omega|$ (many smooth kernels have $\widehat{\kappa}(\omega)$ decreasing in $|\omega|$, yielding slow learning of high frequencies).

Two-sided step target (localized jump). Consider a two-sided step on $\mathbb{T}$ or on a bounded interval, e.g.

$$y(x) = A\left(\mathbb{1}\{|x - x_0| \le \delta\} - \mathbb{1}\{|x - x_1| \le \delta\}\right),$$

which is the difference of two localized bumps ("two-sided" in the sense of two opposite signed supports). Its Fourier coefficients satisfy $|\widehat{y}(\omega)| \asymp |\omega|^{-1}$ for large $|\omega|$ (jump discontinuities generate a $1/\omega$ tail). By (18), each mode decays as $e^{-\widehat{\kappa}(\omega)\tau(t)}$. Hence the time needed to resolve sharp edges is controlled by how quickly $\widehat{\kappa}(\omega)$ decays with $|\omega|$:

- if $\widehat{\kappa}(\omega)$ decays rapidly (very smooth kernels), high frequencies are learned slowly and the step is learned by progressive sharpening;
- if $\widehat{\kappa}(\omega)$ is flatter across a wide band (more localized dictionaries, less smooth kernels), the step "locks in" quickly, producing spike-like localization.

Example: 4-point "two-sided step" dataset $\{0, \frac{1}{3}, \frac{2}{3}, 1\}$ with labels $(0, A, -A, 0)$. Let $x_1 = 0$, $x_2 = \frac{1}{3}$, $x_3 = \frac{2}{3}$, $x_4 = 1$ and $y = (0, A, -A, 0)^\top$. Assume a symmetric distance kernel $K_{ij} = \kappa(|x_i - x_j|)$ (this includes the NNGP kernel in many isotropic setups restricted to a 1D grid). Set

$$k_0 \equiv \kappa(0), \qquad k_1 \equiv \kappa\big(\tfrac{1}{3}\big), \qquad k_2 \equiv \kappa\big(\tfrac{2}{3}\big), \qquad k_3 \equiv \kappa(1),$$

so the Gram matrix is the 4×4 Toeplitz matrix

$$K = \begin{pmatrix} k_0 & k_1 & k_2 & k_3 \\ k_1 & k_0 & k_1 & k_2 \\ k_2 & k_1 & k_0 & k_1 \\ k_3 & k_2 & k_1 & k_0 \end{pmatrix}.$$

Consider the kernel flow (14) with $\hat{y}(0) = 0$, hence $e(0) \equiv \hat{y}(0) - y = -y$. Because K is Toeplitz and y is antisymmetric, the dynamics stay in the 2D antisymmetric subspace

$$e(t) = (a(t),\, b(t),\, -b(t),\, -a(t))^\top, \qquad \text{equivalently} \qquad \hat{y}(t) = (a(t),\, A + b(t),\, -A - b(t),\, -a(t))^\top.$$

In this subspace the error dynamics close as a 2×2 linear system (in effective time $\tau(t) = \int_0^t \eta(s)\, ds$):

$$\partial_\tau \begin{pmatrix} a \\ b \end{pmatrix} = -M \begin{pmatrix} a \\ b \end{pmatrix}, \qquad M = \begin{pmatrix} k_0 - k_3 & k_1 - k_2 \\ k_1 - k_2 & k_0 - k_1 \end{pmatrix}, \qquad (a(0), b(0)) = (0, -A). \tag{19}$$

Write

$$m \equiv \frac{\mathrm{tr}(M)}{2} = k_0 - \frac{k_1 + k_3}{2}, \qquad d \equiv \frac{M_{11} - M_{22}}{2} = \frac{k_1 - k_3}{2}, \qquad o \equiv M_{12} = k_1 - k_2, \qquad s \equiv \sqrt{d^2 + o^2}.$$

Then the solution is explicit:

$$a(t) = A\, e^{-m\tau(t)}\, \frac{o}{s} \sinh\big(s\, \tau(t)\big), \qquad b(t) = -A\, e^{-m\tau(t)} \Big(\cosh\big(s\, \tau(t)\big) + \frac{d}{s} \sinh\big(s\, \tau(t)\big) \Big). \tag{20}$$

In particular, the two nonzero labels are learned through the single scalar $\hat{y}(t, x_2) = A + b(t)$ (and $\hat{y}(t, x_3) = -A - b(t)$), while the endpoints take the coupling-induced values $\hat{y}(t, x_1) = a(t)$ and $\hat{y}(t, x_4) = -a(t)$.

Computing (k_0, k_1, k_2, k_3) explicitly from a closed-form NNGP kernel (ReLU). To obtain concrete numerical rates, one can plug in a closed-form NNGP kernel. Assume:

- $\theta \sim \mathcal{N}(0, I_d)$ and $x, x' \in \mathbb{R}^d$ are embedded/normalized so that $\|x\| = \|x'\| = 1$ and $\rho \equiv \langle x,\, x' \rangle \in [-1, 1]$,
- $\varphi_1(u) = u_+$ (ReLU first-layer features).

Then the NNGP (feature) kernel has the well-known closed form

$$K_{\mu_0}(x, x') = \kappa_{\mathrm{ReLU}}(\rho) \equiv \frac{1}{2\pi} \Big(\sqrt{1 - \rho^2} + (\pi - \arccos \rho)\rho \Big), \qquad \rho = \big\langle x,\, x' \big\rangle\, . \tag{21}$$

If the 1D coordinate $t \in [0, 1]$ is embedded on the unit circle, $x(t) = (\cos(2\pi t), \sin(2\pi t))$, then $\rho(t, t') = \cos(2\pi(t - t'))$ depends only on the distance on the circle. For the grid $\{0, \frac{1}{3}, \frac{2}{3}, 1\}$ one has

$$\rho(0, 1) = 1, \qquad \rho\big(0, \tfrac{1}{3}\big) = \rho\big(0, \tfrac{2}{3}\big) = \cos\big(\tfrac{2\pi}{3}\big) = -\tfrac{1}{2},$$

so $k_0 = k_3 = \kappa_{\mathrm{ReLU}}(1) = \frac{1}{2}$ and $k_1 = k_2 = \kappa_{\mathrm{ReLU}}(-\frac{1}{2})$ with

$$k_1 = k_2 = \frac{1}{2\pi} \Big(\frac{\sqrt{3}}{2} - \frac{\pi}{6} \Big) = \frac{\sqrt{3}}{4\pi} - \frac{1}{12}. \tag{22}$$

In particular $k_1 = k_2$ and $k_0 = k_3$, hence the 2×2 system (19) reduces to the scalar ODE

$$\partial_\tau b(\tau) = -(k_0 - k_1)\, b(\tau), \qquad b(0) = -A,$$

so

$$\hat{y}(t, x_2) = A + b(t) = A\Big(1 - e^{-(k_0-k_1)\tau(t)}\Big), \qquad \hat{y}(t, x_3) = -A - b(t) = -A\Big(1 - e^{-(k_0-k_1)\tau(t)}\Big), \tag{23}$$

with the explicit rate

$$k_0 - k_1 = \frac{1}{2} - \Big(\frac{\sqrt{3}}{4\pi} - \frac{1}{12}\Big) = \frac{7}{12} - \frac{\sqrt{3}}{4\pi}.$$

Lemma 9 (Exact learning curve for the two-sided step on the 4-point grid in the ReLU NNGP kernel regime)**.** *In the 4-point dataset $x_1 = 0$, $x_2 = \frac{1}{3}$, $x_3 = \frac{2}{3}$, $x_4 = 1$ with $y = (0, A, -A, 0)^\top$, assume the fixed-kernel gradient flow* (14) *with $\hat{y}(0) = 0$ and the ReLU NNGP kernel* (21) *under the unit-circle embedding $x(t) = (\cos(2\pi t), \sin(2\pi t))$. Then the predictions at the two nonzero-label points satisfy the explicit learning curves*

$$\hat{y}(t, x_2) = A\Big(1 - e^{-\lambda\,\tau(t)}\Big), \qquad \hat{y}(t, x_3) = -A\Big(1 - e^{-\lambda\,\tau(t)}\Big), \qquad \lambda = \frac{7}{12} - \frac{\sqrt{3}}{4\pi},$$

and $\hat{y}(t, x_1) = \hat{y}(t, x_4) = 0$ for all t. In particular, the (normalized) step amplitude

$$g(t) \equiv \frac{\hat{y}(t, x_2)}{A} = 1 - e^{-\lambda\,\tau(t)}$$

increases monotonically from 0 to 1 at an exponential rate controlled by λ.

Remark 8 (How to view this as a "curve of f_k" in the spike dictionary picture)**.** *In Step 5.2, in a diagonal-dominant regime the channel functions are superpositions of kernel bumps. On the 4-point step dataset, the target is supported on the two interior points $\{x_2, x_3\}$. If one is in a regime where the two channels specialize so that channel 1 carries the positive spike and channel 2 carries the negative spike, a natural idealized representation is*

$$f_1(t, x) \approx A\, g(t)\, \frac{K_{\mu_0}(x, x_2)}{K_{\mu_0}(x_2, x_2)}, \qquad f_2(t, x) \approx -A\, g(t)\, \frac{K_{\mu_0}(x, x_3)}{K_{\mu_0}(x_3, x_3)},$$

with the same scalar amplitude curve $g(t) = 1 - e^{-\lambda\tau(t)}$. For the ReLU NNGP kernel under the unit-circle embedding, this becomes an explicit bump shape on $t \in [0, 1]$: for $x = x(t)$ and $x_p = x(t_p)$,

$$K_{\mu_0}(x(t), x(t_p)) = \kappa_{\mathrm{ReLU}}(\cos(2\pi(t - t_p))),$$

so the entire shape of $f_1(t, \cdot)$ and $f_2(t, \cdot)$ is given in closed form by κ_{ReLU} and the scalar amplitude $g(t)$. The remaining mathematical task (handled by Step 4.5–Step 5.3) is to justify this specialization from the nonlinear mean-field dynamics, rather than postulating it.

Remark 9 (High-dimensional "almost diagonal" limit gives an even simpler closed form)**.** *If off-diagonal kernel entries are approximately equal (a common high-dimensional concentration pattern after centering), $k_1 \approx k_2 \approx k_3 \approx k_{\mathrm{off}}$, then $o \approx 0$ and $d \approx 0$, hence $a(t) \approx 0$ and*

$$\hat{y}(t, x_2) \approx A\big(1 - e^{-(k_0-k_{\mathrm{off}})\tau(t)}\big), \qquad \hat{y}(t, x_3) \approx -A\big(1 - e^{-(k_0-k_{\mathrm{off}})\tau(t)}\big),$$

so the "frequency" mode $(0, 1, -1, 0)$ is learned at rate $k_0 - k_{\mathrm{off}}$. If moreover the features are mean-zero / centered so that $k_{\mathrm{off}} \approx 0$, then the four points essentially decouple and the learning rate is $\approx k_0$.

Remark 10 (What is "fully solved" and what remains model-specific)**.** *The formulas* (15)–(18) *are an exact closed-form solution* once *one commits to a fixed kernel (NTK regime). The model-specific work is to justify that, in the MMNN low-rank mean-field dynamics, the relevant Gram operator is close to a fixed* K_{μ_0} *(or to quantify how gating makes it time-dependent). In the diagonal-dominant regime* $K \approx \mathrm{diag}(K_{11}, \dots, K_{nn})$*,* (15) *reduces to* n *independent scalar ODEs and step-like targets behave as collections of nearly independent "spikes".*

4.10 Step 5.2: solving the mean-field w_1-PDE explicitly for finitely-supported data (and reducing spike learning to scalar ODEs)

We now go back from the fixed-kernel linearization (Step 5.1) to the actual mean-field equations for (w_1, w_2). The key point is that the w_1 evolution is *linear* in feature space once $B_k(t; \cdot)$ and the residual signal are fixed, so for finitely-supported data one can solve $\bar{w}_{1,k}$ and hence f_k *exactly* as a superposition of kernel bumps.

Finite-support data and scalar coefficients. Assume the input marginal is supported on finitely many points $x^{(1)}, \dots, x^{(m)}$:

$$X \in \{x^{(1)}, \dots, x^{(m)}\}, \qquad \mathbb{P}(X = x^{(p)}) = \pi_p, \qquad p = 1, \dots, m,$$

with corresponding targets $y^{(p)}$. For each location $x^{(p)}$, write

$$d_p(t) \equiv d_L\big((x^{(p)}, y^{(p)}); W(t)\big), \qquad B_{k,p}(t) \equiv B_k(t; x^{(p)}).$$

Then (1) becomes the explicit ODE (pointwise in θ)

$$\partial_t \bar{w}_{1,k}(t, \theta) = -\xi_1(t) \sum_{p=1}^{m} \pi_p \, d_p(t) \, B_{k,p}(t) \, \varphi_1(\big\langle \theta, \, x^{(p)} \big\rangle), \qquad k = 1, \dots, r. \tag{24}$$

Integrating (24) in time yields the closed form

$$\bar{w}_{1,k}(t, \theta) = \bar{w}_{1,k}(0, \theta) - \sum_{p=1}^{m} \varphi_1(\big\langle \theta, \, x^{(p)} \big\rangle) \, \Gamma_{k,p}(t), \qquad \Gamma_{k,p}(t) \equiv \int_0^t \xi_1(s) \, \pi_p \, d_p(s) \, B_{k,p}(s) \, ds. \tag{25}$$

Plugging (25) into the definition of f_k gives an explicit superposition in input space:

$$f_k(t, x) = f_k(0, x) - \sum_{p=1}^{m} \Gamma_{k,p}(t) \, K_{\mu_0}(x, x^{(p)}), \qquad k = 1, \dots, r. \tag{26}$$

Thus, in finitely-supported settings, the *spike shape* of each channel is completely controlled by the dictionary kernel K_{μ_0}, while all learning dynamics reduce to the scalar coefficients $\Gamma_{k,p}(t)$.

One-spike reduction (exact spike shape). If $m = 1$ with $x^{(1)} = x_\star$ (a single spike location), then (26) becomes

$$f_k(t, x) = f_k(0, x) - \Gamma_{k,1}(t) \, K_{\mu_0}(x, x_\star).$$

In particular, if $f_k(0, \cdot) \equiv 0$ then $f_k(t, \cdot)$ is *exactly* a kernel bump at $x_\star$ for all t, and "spike learning" reduces to solving $\Gamma_{k,1}(t)$ (equivalently $f_k(t, x_\star)$). This makes precise why Step 5 depends on the localization of K_{μ_0}: the dynamics can only create spikes to the extent that $K_{\mu_0}(x, x_\star)$ is itself localized.

Two-sided step as a two-spike reduction. A two-sided step can be idealized as two opposite-signed locations, $m = 2$, e.g. $y^{(1)} = +A$ at $x^{(1)} = x_0$ and $y^{(2)} = -A$ at $x^{(2)} = x_1$. Then each channel has the exact representation

$$f_k(t,x) = f_k(0,x) - \Gamma_{k,1}(t)\, K_{\mu_0}(x,x_0) - \Gamma_{k,2}(t)\, K_{\mu_0}(x,x_1).$$

If K_{μ_0} is diagonal-dominant on $\{x_0, x_1\}$ (i.e. $|K_{\mu_0}(x_0,x_1)| \ll K_{\mu_0}(x_0,x_0)$), the two coefficients $(\Gamma_{k,1}, \Gamma_{k,2})$ are only weakly coupled through the residuals, and one obtains two almost-independent "local" spike dynamics (one near x_0 and one near x_1).

How to read "instability at initialization" in this formulation. In this finite-support reduction, "initialization is unstable" toward specialization at a location $x^{(p)}$ means: starting from small but unequal coefficients $\Gamma_{k,p}(0)$ across k, the log-ratios

$$R_{k\ell}(t,x^{(p)}) = \log \frac{|f_k(t,x^{(p)})| + \varepsilon}{|f_\ell(t,x^{(p)})| + \varepsilon}$$

grow because the per-capita gains (Step 4.5) generate a persistent gap. In the exact one-sparse spike regime of Lemma 5, this instability is extreme: all non-dominant channels satisfy $B_k(t) = 0$ and hence $\Gamma_{k,p}(t) \equiv 0$, while the dominant channel has $B_j(t) > 0$ and $\Gamma_{j,p}(t)$ grows in the descent direction as long as the residual has the appropriate sign. The remaining (hard) step beyond this exact regime is a stability estimate: if one channel is only *approximately* dominant, then B_k (hence $\Gamma_{k,p}$) is small but nonzero for $k \neq j$, yielding quantitative log-ratio growth rather than exact decoupling.

Case $r = 2$, two-spike setting, and "slight dominance" at one spike. Take $r = 2$ channels and $m = 2$ spike locations $x^{(1)} = x_0$ and $x^{(2)} = x_1$ with opposite targets (two-sided step). Assume a diagonal-dominant local regime so that $K_{\mu_0}(x_0,x_1)$ is negligible compared to $K_{\mu_0}(x_0,x_0)$ and $K_{\mu_0}(x_1,x_1)$. If moreover $f_k(0,\cdot) \equiv 0$, then evaluating (26) at x_0 gives the local approximation

$$f_k(t,x_0) \approx -\Gamma_{k,1}(t)\, K_{\mu_0}(x_0,x_0), \qquad k = 1,2, \tag{27}$$

so "channel 1 is slightly dominant at the spike x_0" is the quantitative condition

$$|f_1(t,x_0)| > |f_2(t,x_0)| \qquad \Longleftrightarrow \qquad |\Gamma_{1,1}(t)| > |\Gamma_{2,1}(t)|$$

in the local approximation (27). To conclude that this dominance is self-reinforcing, one needs a *stability* form of the one-sparse mechanism: the backprop signals satisfy a bound of the form $|B_{2,1}(t)| \ll |B_{1,1}(t)|$ whenever $|f_2(t,x_0)| \ll |f_1(t,x_0)|$.

Lemma 10 (A local log-ratio growth criterion for $r = 2$)**.** *Fix the spike location x_0 and assume the local reduction* (11) *holds there:*

$$\partial_t f_k(t,x_0) = -\xi_1(t)\, K_{\mu_0}(x_0,x_0)\, d_0(t)\, B_{k,1}(t), \qquad k = 1,2,$$

where $d_0(t)$ is the residual signal at x_0 and $B_{k,1}(t) = B_k(t;x_0)$. Assume there exists a constant $\rho \in [0,1)$ such that on a time interval I,

$$\big|B_{2,1}(t)\big| \le \rho\, \frac{|f_2(t,x_0)| + \varepsilon}{|f_1(t,x_0)| + \varepsilon}\, \big|B_{1,1}(t)\big| \qquad \textit{for all } t \in I. \tag{28}$$

If moreover $-d_0(t) \geq 0$ and $B_{1,1}(t)$ has the same sign as $f_1(t, x_0)$ on I, then the log-ratio

$$R_{12}(t, x_0) = \log \frac{|f_1(t, x_0)| + \varepsilon}{|f_2(t, x_0)| + \varepsilon}$$

satisfies $\partial_t R_{12}(t, x_0) \geq (1 - \rho)\, \xi_1(t)\, K_{\mu_0}(x_0, x_0)\, (-d_0(t))\, \frac{|B_{1,1}(t)|}{|f_1(t,x_0)|+\varepsilon} \geq 0$ for a.e. $t \in I$. In particular, any strict dominance at $t = t_0 \in I$ cannot be lost on I and is amplified whenever $|B_{1,1}|$ is not too small.

Proof. By the exact identity (10) at $x = x_0$,

$$\partial_t R_{12}(t, x_0) = \frac{\operatorname{sign}(f_1(t, x_0))\, \partial_t f_1(t, x_0)}{|f_1(t, x_0)| + \varepsilon} - \frac{\operatorname{sign}(f_2(t, x_0))\, \partial_t f_2(t, x_0)}{|f_2(t, x_0)| + \varepsilon}.$$

Insert the local dynamics and use $-d_0(t) \geq 0$:

$$\partial_t R_{12}(t, x_0) = \xi_1(t)\, K_{\mu_0}(x_0, x_0)\, (-d_0(t)) \Big(\frac{\operatorname{sign}(f_1) B_{1,1}}{|f_1| + \varepsilon} - \frac{\operatorname{sign}(f_2) B_{2,1}}{|f_2| + \varepsilon} \Big).$$

By the sign assumption on $B_{1,1}$, $\operatorname{sign}(f_1) B_{1,1} = |B_{1,1}|$. Moreover $-\operatorname{sign}(f_2) B_{2,1} \geq -|B_{2,1}|$. Using (28) yields

$$\frac{|B_{1,1}|}{|f_1| + \varepsilon} - \frac{\operatorname{sign}(f_2) B_{2,1}}{|f_2| + \varepsilon} \geq \frac{|B_{1,1}|}{|f_1| + \varepsilon} - \frac{|B_{2,1}|}{|f_2| + \varepsilon} \geq (1 - \rho) \frac{|B_{1,1}|}{|f_1| + \varepsilon},$$

which proves the claimed lower bound for $\partial_t R_{12}$. □

Lemma 11 (Two-sided step (two spikes): log-ratio growth at each spike under local decoupling)**.** *Assume the two-sided step setting of Step 5.2 with $m = 2$ spike locations $x^{(1)} = x_0$ and $x^{(2)} = x_1$, and take $r = 2$ channels. Assume a diagonal-dominant regime so that the local reductions hold at each spike: for $p \in \{0, 1\}$ and $k \in \{1, 2\}$,*

$$\partial_t f_k(t, x_p) = -\xi_1(t)\, K_{\mu_0}(x_p, x_p)\, d_p(t)\, B_{k,p+1}(t),$$

where $d_p(t)$ is the residual signal at x_p and $B_{k,p+1}(t) = B_k(t; x^{(p+1)})$. Suppose that on a time interval I:

- *$-d_0(t) \geq 0$ and $B_{1,1}(t)$ has the same sign as $f_1(t, x_0)$, and there exists $\rho_0 \in [0, 1)$ such that*

$$|B_{2,1}(t)| \leq \rho_0 \frac{|f_2(t, x_0)| + \varepsilon}{|f_1(t, x_0)| + \varepsilon} |B_{1,1}(t)|,$$

- *$-d_1(t) \geq 0$ and $B_{2,2}(t)$ has the same sign as $f_2(t, x_1)$, and there exists $\rho_1 \in [0, 1)$ such that*

$$|B_{1,2}(t)| \leq \rho_1 \frac{|f_1(t, x_1)| + \varepsilon}{|f_2(t, x_1)| + \varepsilon} |B_{2,2}(t)|.$$

Then the log-ratios at the two spikes satisfy, for a.e. $t \in I$,

$$\partial_t R_{12}(t, x_0) \geq (1 - \rho_0)\, \xi_1(t)\, K_{\mu_0}(x_0, x_0)\, (-d_0(t))\, \frac{|B_{1,1}(t)|}{|f_1(t, x_0)| + \varepsilon} \geq 0,$$

and

$$\partial_t R_{21}(t, x_1) \geq (1 - \rho_1)\, \xi_1(t)\, K_{\mu_0}(x_1, x_1)\, (-d_1(t))\, \frac{|B_{2,2}(t)|}{|f_2(t, x_1)| + \varepsilon} \geq 0,$$

where $R_{21}(t, x_1) = \log \frac{|f_2(t,x_1)|+\varepsilon}{|f_1(t,x_1)|+\varepsilon}$. In particular, any initial strict dominance of channel 1 at x_0 and of channel 2 at x_1 cannot be lost on I and is amplified whenever the corresponding $|B|$ terms are not too small.

Remark 11 (Why this is the "step setup" version of Lemma 10)**.** *Lemma 10 gives a one-spike sufficient condition for winner-take-all (ratio growth) at a single location. Lemma 11 is the direct two-spike specialization: under local decoupling, one applies the same log-ratio calculus at both spike locations. The remaining model-specific work is therefore reduced to verifying the stability-type inequalities on B (the "$\rho < 1$" conditions) from the gating dynamics and the mixing structure.*

Remark 12 (How (28) can arise from a half-space gate)**.** *Condition* (28) *is an abstract "approximate one-sparse stability" inequality. In the random-sign mixing picture, it is expected to hold when the gate $\mathbb{1}\{\langle L_{C_2}, f(t,x_0)\rangle > 0\}$ is largely determined by the dominant coordinate (here channel 1), so that the effective selection of c_2-indices creates a strong conditional bias for $B_{1,1}$ but only a weak (order $|f_2|/|f_1|$) bias for $B_{2,1}$. Making this fully rigorous requires quantitative control of the dependence of $w_2(t, C_2)$ on the small coordinate through* (8)*; in the exact one-sparse regime of Lemma 5 one has $\rho = 0$.*

4.11 Step 5.3: the full local $r = 2$ spike dynamics for (f_1, f_2), per-capita gains, and what can (and cannot) be "solved"

This subsection writes the complete closed system at a spike location in the local regime and clarifies which structural assumptions allow an explicit solution or yield (non-)specialization.

Local two-channel dynamics at one spike location. Fix a spike location x_0 and assume a diagonal-dominant local regime so that $K_{\mu_0}(x_0, X) \approx K_{\mu_0}(x_0, x_0)\mathbb{1}\{X = x_0\}$. Write

$$u_1(t) \equiv f_1(t, x_0), \qquad u_2(t) \equiv f_2(t, x_0), \qquad u(t) = (u_1(t), u_2(t)) \in \mathbb{R}^2,$$

and let $L_{c_2} = (L_{c_2,1}, L_{c_2,2}) \in \mathbb{R}^2$ be the frozen mixing vector for index c_2. Define the preactivation and gate at x_0:

$$S(t, c_2) \equiv \langle L_{c_2},\, u(t)\rangle, \qquad G(t, c_2) \equiv \mathbb{1}\{S(t, c_2) > 0\}.$$

For ReLU top activation $\varphi_2(u) = u_+$ and square loss at (x_0, y_0), the local residual is $d_0(t) \equiv \hat{y}(t, x_0) - y_0$ with

$$\hat{y}(t, x_0) = \mathbb{E}_{C_2}\big[w_2(t, C_2)\, S(t, C_2)\, G(t, C_2)\big]. \tag{29}$$

The backprop signals are

$$B_k(t; x_0) = \mathbb{E}_{C_2}\big[L_{C_2,k}\, w_2(t, C_2)\, G(t, C_2)\big], \qquad k = 1, 2. \tag{30}$$

Then the (local) channel dynamics are

$$\partial_t u_k(t) = -\xi_1(t)\, K_{\mu_0}(x_0, x_0)\, d_0(t)\, B_k(t; x_0), \qquad k = 1, 2, \tag{31}$$

and the top-layer mean-field dynamics are (finite-support reduction with one point)

$$\partial_t w_2(t, c_2) = -\xi_2(t)\, d_0(t)\, S(t, c_2)\, G(t, c_2). \tag{32}$$

Equations (29)–(32) are the complete closed ODE system for $(u(t), w_2(t, \cdot))$ at one spike location.

Per-capita gains and log-ratio dynamics. Define the log-ratio at x_0,

$$R_{12}(t, x_0) = \log \frac{|u_1(t)| + \varepsilon}{|u_2(t)| + \varepsilon}.$$

Combining (10) with (31) yields the exact identity (a.e. in t)

$$\partial_t R_{12}(t, x_0) = \xi_1(t)\, K_{\mu_0}(x_0, x_0)\, (-d_0(t)) \Big(\frac{\operatorname{sign}(u_1(t))\, B_1(t; x_0)}{|u_1(t)| + \varepsilon} - \frac{\operatorname{sign}(u_2(t))\, B_2(t; x_0)}{|u_2(t)| + \varepsilon} \Big). \tag{33}$$

Thus the whole winner-take-all question at a spike reduces to comparing the *per-capita gains*

$$\gamma_k(t, x_0) \equiv \xi_1(t)\, K_{\mu_0}(x_0, x_0)\, (-d_0(t))\, \frac{\operatorname{sign}(u_k(t))\, B_k(t; x_0)}{|u_k(t)| + \varepsilon}, \qquad k = 1, 2.$$

What is explicitly solvable.

- *Exact one-sparse manifold.* If $u_2(0) = 0$ and the mixing distribution is such that $B_2(t; x_0) = 0$ whenever $u_2(t) = 0$ (as in Lemma 5), then $u_2(t) \equiv 0$ and u_1 reduces to the scalar dynamics already solved in Lemma 5 and Lemma 7.
- *Fixed-kernel linearization.* If one replaces (B_1, B_2) by a linear function of u frozen at initialization, one recovers the matrix exponential solution of Step 5.1.

A useful warning: isotropic Gaussian mixing forces $B(t; x_0)$ to be parallel to $u(t)$ (no ratio dynamics). The half-space mechanism alone does *not* automatically create specialization when the mixing distribution is isotropic. The following lemma makes this precise.

Lemma 12 (Isotropic Gaussian mixing implies B is colinear with u)**.** *Assume $L_{C_2} \sim \mathcal{N}(0, I_2)$ and that $w_2(t, C_2)$ is a measurable function of $S(t, C_2) = \langle L_{C_2}, u(t)\rangle$ only, i.e. $w_2(t, C_2) = g_t(S(t, C_2))$ for some function $g_t : \mathbb{R} \to \mathbb{R}$. Then for any $u(t) \neq 0$,*

$$\big(B_1(t; x_0), B_2(t; x_0)\big) = c(t)\, u(t)$$

for a scalar coefficient $c(t)$ depending on $u(t)$ only through $\|u(t)\|$ and on g_t through $\mathbb{E}[S g_t(S) \mathbb{1}\{S > 0\}]$. Consequently, if $u_1(t), u_2(t)$ have the same sign, then the bracket in (33) *vanishes and $R_{12}(t, x_0)$ is constant (no winner-take-all).*

Proof. Let $L \sim \mathcal{N}(0, I_2)$ and $S = \langle L, u\rangle$. By Gaussian regression, $\mathbb{E}[L \mid S] = \frac{u}{\|u\|^2} S$. Therefore

$$\mathbb{E}\big[L\, g_t(S)\, \mathbb{1}\{S > 0\}\big] = \mathbb{E}\big[\mathbb{E}[L \mid S]\, g_t(S)\, \mathbb{1}\{S > 0\}\big] = \frac{u}{\|u\|^2} \mathbb{E}\big[S g_t(S)\, \mathbb{1}\{S > 0\}\big],$$

which proves colinearity. If u_1, u_2 have the same sign then $\operatorname{sign}(u_k) B_k = |B_k|$ and $|B_k|/(|u_k| + \varepsilon)$ is proportional to $|u_k|/(|u_k| + \varepsilon)$ with the same proportionality constant, so the difference in (33) cancels. ☐

Remark 13 (What this means for your experiments)**.** *Lemma 12 explains why proving specialization requires more than "half-space gating + symmetry": one needs either (i) non-isotropic / structured mixing L (e.g. dictionary-like rows), or (ii) multi-spike interactions (different gates at different x reshape w_2 in a way that breaks the $w_2 = g(S)$ symmetry at a single location), or (iii) higher-layer structure beyond the one-spike reduction. In such settings, the stability hypothesis* (28) *becomes plausible and yields strict log-ratio growth via Lemma 10.*

4.12 Step 5.4: seeding strict dominance by the first SGD step (channel-resolved NTK/linearization)

The previous steps show that *if* one has strict dominance at each spike location and a stability bound of the form (28), then winner-take-all amplification follows. This subsection explains a complementary (and often simpler) mechanism to obtain the *initial* strict dominance: the very first stochastic gradient step (finite-width, random initialization) typically produces a small but *random* channel imbalance at each spike. One then bootstraps: first-step imbalance $\Rightarrow$ log-ratio becomes positive, and afterwards the continuous-time arguments apply.

Discrete-time first step in the two-spike reduction. Consider the two-spike (two-sided step) dataset with $x^{(1)} = x_0$, $x^{(2)} = x_1$ and $y^{(1)} = +A$, $y^{(2)} = -A$, and take $r = 2$ channels. Assume a local/diagonal-dominant regime so that the evolution at each spike is governed primarily by the diagonal kernel values $K_{\mu_0}(x_p, x_p)$. Assume also that the initial prediction is close to 0 at both spikes, so the initial residuals satisfy

$$d_0(0) \approx -A, \qquad d_1(0) \approx +A,$$

for square loss (underfitting at x_0 and overfitting at x_1 relative to 0).

Let $\eta > 0$ be a (small) SGD step size and consider the first-step update of the channel values at the spikes under a channel-resolved linearization (freeze the kernel K_{μ_0} and the backprop signals $B_{k,p}(0)$ at initialization):

$$f_k^+(x_p) \equiv f_k(0, x_p) - \eta\, \xi_1(0)\, \pi_p\, d_p(0)\, B_{k,p}(0)\, K_{\mu_0}(x_p, x_p), \qquad k = 1, 2, \quad p \in \{0, 1\}, \tag{34}$$

where $\pi_p = \mathbb{P}(X = x_p)$ and $B_{k,p}(0) = B_k(0; x^{(p+1)})$. The precise update rule depends on the optimizer/minibatch, but (34) captures the leading-order effect: the first step creates channel amplitudes proportional to the random initialization of $B_{k,p}(0)$.

Lemma 13 (First-step strict dominance occurs with constant probability under exchangeable initialization)**.** *Assume that at initialization the random variables* $(|B_{1,1}(0)|, |B_{2,1}(0)|)$ *are i.i.d. with a continuous distribution, and likewise* $(|B_{1,2}(0)|, |B_{2,2}(0)|)$ *are i.i.d., independent of the first pair. Assume* $f_k(0, x_p) = 0$ *and that* (34) *holds with* $K_{\mu_0}(x_p, x_p) > 0$ *and* $\pi_p > 0$*. Then:*

- *at* x_0*,* $\mathbb{P}\big(|f_1^+(x_0)| > |f_2^+(x_0)|\big) = \frac{1}{2}$*;*
- *at* x_1*,* $\mathbb{P}\big(|f_2^+(x_1)| > |f_1^+(x_1)|\big) = \frac{1}{2}$*;*
- *consequently, the "correct assignment" event*

$$\mathcal{E} \equiv \{|f_1^+(x_0)| > |f_2^+(x_0)|\} \cap \{|f_2^+(x_1)| > |f_1^+(x_1)|\}$$

  has probability $\mathbb{P}(\mathcal{E}) = \frac{1}{4}$*.*

Moreover, on $\mathcal{E}$ *the initial log-ratios are strictly positive:*

$$R_{12}(0^+, x_0) = \log \frac{|f_1^+(x_0)| + \varepsilon}{|f_2^+(x_0)| + \varepsilon} > 0, \qquad R_{21}(0^+, x_1) = \log \frac{|f_2^+(x_1)| + \varepsilon}{|f_1^+(x_1)| + \varepsilon} > 0.$$

Proof. From (34) and $f_k(0, x_p) = 0$, one has

$$|f_k^+(x_p)| = \eta\, \xi_1(0)\, \pi_p\, |d_p(0)|\, K_{\mu_0}(x_p, x_p)\, |B_{k,p}(0)|.$$

The prefactor does not depend on k, so comparing $|f_1^+(x_p)|$ and $|f_2^+(x_p)|$ is equivalent to comparing $|B_{1,p}(0)|$ and $|B_{2,p}(0)|$. By i.i.d. continuity, $\mathbb{P}(|B_{1,p}(0)| > |B_{2,p}(0)|) = \frac{1}{2}$. Independence across spikes gives $\mathbb{P}(\mathcal{E}) = \frac{1}{4}$. Finally, strict inequalities imply the log-ratios at 0^+ are positive for any $\varepsilon \geq 0$. $\square$

Remark 14 (How this plugs into the continuous-time winner-take-all argument). *Lemma 13 is only a* seeding *result: it produces an initial strict dominance with constant probability from exchangeable random initialization. To conclude that the dominance is* amplified *over time, one then proves a stability bound of the form* (28) *(or its two-spike version in Lemma 11), which yields monotone log-ratio growth. This is exactly the bootstrap structure: first-step symmetry breaking ⇒ ratios become positive ⇒ stability implies ratios keep increasing.*

4.13 Step 6: what is realistically provable first

A plausible first theorem is a *symmetry-breaking / specialization* result: under sign-coherence and a diagonal-dominant kernel assumption, the symmetric state where channels are comparable becomes unstable and the dynamics drive $s_k(t, x)$ toward $\{0, 1\}$ (one dominant channel) for most x. This already captures the observed "different channels learn different slots" effect, and it is robust to small perturbations (off-diagonal kernel terms) via Grönwall and stability estimates.

5 Layer-wise spike sharpening and frequency growth across depth

This section targets the statement you want to prove. It does *not* start by defining a Fourier cutoff and proving a generic localization inequality; instead it isolates the mean-field mechanism that can generate *geometric harmonic growth* when fitting a highly oscillatory target.

5.1 What must be proved (informal statement)

When the target is highly oscillatory, the trained network exhibits layerwise partial functions that are increasingly oscillatory and increasingly spike-like. The proof goal is to show that, in a sign-coherent regime, the mean-field gradient-flow update acts (after averaging over the gate) like a *sharpening operator* with gain strictly larger than 1, so that a characteristic spatial scale shrinks geometrically across depth, yielding geometric growth of accessible frequencies.

5.2 A minimal 1D mean-field sharpening model

Work on a 1D periodic domain $\mathbb{T}$. Let $u^{(\ell)}(t, \cdot)$ denote an effective *pre-ReLU* scalar field at depth ℓ (e.g. a dominant low-rank channel or a signed linear combination of channels). Define the *post-ReLU* field $v^{(\ell)}(t, x) = \max\{u^{(\ell)}(t, x), 0\}$. The empirical phenomenon is that if $u^{(\ell)}$ has many sign changes, then $v^{(\ell)}$ is supported on many short intervals, producing a collection of localized spikes.

The mean-field gradient flow updates $u^{(\ell)}$ through a gated expectation of the schematic form

$$\partial_t u^{(\ell)}(t, \cdot) \approx -\mathcal{K}_t^{(\ell)}\Big(d(t, \cdot)\, \mathcal{G}_t^{(\ell)}[u^{(\ell)}(t, \cdot)] \Big), \tag{35}$$

where $\mathcal{K}_t^{(\ell)}$ is a (layer-dependent) integral operator induced by the feature measure at depth ℓ, $d(t, \cdot)$ is the residual, and $\mathcal{G}_t^{(\ell)}$ is a *gate-induced* multiplication operator (coming from $\mathbb{1}\{S > 0\}$ and its correlations with the low-rank coordinates). The point is that $\mathcal{G}_t^{(\ell)}$ is *not* neutral: it biases updates toward the active set and can increase the effective slope/scale.

5.3 Half-space constants and why variance 2 matters

The basic half-space constants are explicit. If $G \sim \mathcal{N}(0, 2)$ then

$$\mathbb{E}|G| = \frac{2}{\sqrt{\pi}} > 1, \qquad \mathbb{E}\big[G\, \mathbb{1}\{G > 0\}\big] = \frac{1}{\sqrt{\pi}}. \tag{36}$$

In the one-sparse / sign-coherent regimes used earlier, the conditional expectations over the gate reduce many vector-valued expressions to one-dimensional half-space averages. Equation (36) is exactly the place where the "variance 2" parameterization creates a gain constant > 1.

5.4 A provable geometric harmonic-growth principle (template)

We now isolate a statement that captures the mechanism: *gate bias increases an effective slope parameter*, and slope growth implies shrinking spike widths, hence higher resolvable frequencies. To avoid introducing new Fourier notation, we quantify oscillation by the number of sign changes and by a characteristic spike width.

Definition 2 (Spike-width proxy)**.** *For a nonnegative function $v : \mathbb{T} \to \mathbb{R}$ that is supported on a union of intervals, define its spike width proxy by*

$$h(v) \equiv \frac{\mathrm{Leb}\big(\mathrm{supp}(v)\big)}{N(v)},$$

where $N(v)$ is the number of connected components of $\mathrm{supp}(v)$ and Leb denotes Lebesgue measure on $\mathbb{T}$. When $v = u_+$ for an oscillatory u, $N(v)$ is proportional to the number of positive excursions, and $h(v)$ measures the typical width of a spike.

Assumption 2 (Mean-field gate sharpening across depth)**.** *There exist deterministic constants $c_\ell > 1$ such that, on a regime where a single dominant channel controls the gate, the depth-ℓ to depth-$(\ell + 1)$ effective map satisfies the following implication: if $u^{(\ell)}(t, \cdot)$ has oscillation scale h_ℓ in the sense that $v^{(\ell)}(t, \cdot) = (u^{(\ell)}(t, \cdot))_+$ has spike width $h(v^{(\ell)}(t, \cdot)) \approx h_\ell$, then $v^{(\ell+1)}(t, \cdot)$ satisfies*

$$h\big(v^{(\ell+1)}(t,\cdot)\big) \;\lesssim\; \frac{1}{c_\ell}\, h\big(v^{(\ell)}(t,\cdot)\big), \qquad \textit{with} \qquad c_\ell \approx \mathbb{E}|G|. \tag{37}$$

Lemma 14 (Geometric shrinkage implies geometric frequency growth)**.** *Assume* (37) *holds along depths $1, \dots, L-1$ and that $c_\ell \geq c > 1$ uniformly. Then the typical spike width satisfies $h(v^{(\ell)}(t, \cdot)) \lesssim c^{-(\ell-1)} h(v^{(1)}(t, \cdot))$. In particular, the number of spikes satisfies*

$$N\big(v^{(\ell)}(t,\cdot)\big) \;\gtrsim\; c^{\ell-1}\, N\big(v^{(1)}(t,\cdot)\big) \quad \textit{whenever} \quad \mathrm{Leb}(\mathrm{supp}(v^{(\ell)}))\ \textit{stays bounded below.}$$

Thus, as depth increases, the representation necessarily contains structure at spatial scales of order $c^{-(\ell-1)}$, i.e. at frequencies of order $c^{\ell-1}$.

Proof. Iterate (37) to obtain geometric shrinkage of h. Since $h(v) = \mathrm{Leb}(\mathrm{supp}(v))/N(v)$ by Definition 2, a lower bound on $\mathrm{Leb}(\mathrm{supp}(v^{(\ell)}))$ converts width shrinkage into growth of the number of connected components. $\square$

Remark 15 (What remains to turn this into a theorem for your MMNN mean-field PDE)**.** *Lemma 14 makes explicit what must be proved: the core task is to justify Assumption 2 from the actual mean-field equations. Concretely, one needs to show that, when fitting an oscillatory target, the gate-induced operator $\mathcal{G}_t^{(\ell)}$ in* (35) *increases an effective slope parameter by a factor comparable to $\mathbb{E}|G|$ (with $G \sim \mathcal{N}(0, 2)$), and that this translates into the spike-width recursion* (37)*. This is where the earlier sign-coherence / half-space bias lemmas are used: they reduce the gate statistics to one-dimensional half-space expectations, producing the gain constant in* (36)*.*

5.5 Dictionary viewpoint and the $2L$ oscillation pattern (why depth is the right lens)

Your observation can be rephrased in a way that is much closer to existing ReLU theory: in 1D, deep ReLU networks represent splines (or spline-like objects), and oscillations are created by the accumulation of *gates* (kinks / cut points). In the mean-field parameterization, the first layer already provides a universal dictionary; depth then controls how many times the input domain is *cut* by gates, which is what creates many alternating spikes. Empirically, in your MMNN low-rank setting, this appears to follow a *linear* pattern: the "L-th block has about $2L$ oscillations" (as opposed to an exponential 2^L worst-case expressivity bound).

First layer as a universal dictionary. On 1D inputs, the ridge dictionary $x \mapsto (x-b)_+$ is universal for piecewise-linear splines. In the mean-field limit, choosing a signed measure over biases b and slopes is exactly choosing a (possibly very large) spline dictionary expansion. Thus the "rank-r components are basis functions" intuition is correct: at each layer, a small number of channel mixtures selects and reweights a huge underlying dictionary.

Depth can create many breakpoints, but training may realize only linear growth. A standard expressivity fact in 1D is that, for generic fully-connected ReLU networks, the number of linear pieces can grow fast with depth (often exponentially in worst-case constructions). However, in your *rank-constrained* low-rank/MMNN blocks, the effective mixing is bottlenecked (small r, sign-coherent gates, locality), and the observed regime is much closer to an *additive* creation of oscillations: each additional block contributes only $O(1)$ new oscillations, leading to a pattern of order $2L$.

Definition 3 (Oscillation count)**.** *For a continuous piecewise-linear function $f : \mathbb{T} \to \mathbb{R}$, let $\mathsf{osc}(f)$ denote the number of maximal intervals on which f is affine (equivalently, one plus the number of breakpoints). This quantity controls the number of sign changes and the number of spikes after thresholding.*

Lemma 15 (1D deep ReLU networks are splines with exponentially many pieces)**.** *Consider a standard fully-connected ReLU network on 1D input $x \in \mathbb{T}$, of width m and depth L (i.e. L hidden layers), with scalar output. Then the realized function is continuous piecewise-linear and satisfies*

$$\mathsf{osc}(f) \ \le \ C\, m^L$$

for a universal constant C (e.g. one may take $C = 2$).

Proof. Each hidden unit computes $\sigma(g(x))$ where g is a linear combination of previous-layer outputs. A linear combination of piecewise-linear functions is piecewise-linear with breakpoints contained in the union of the previous breakpoints, hence the number of pieces adds across units. Applying $\sigma(\cdot) = \max\{\cdot, 0\}$ cannot create more breakpoints than the input piecewise-linear function has (it only clips on subsets of existing affine pieces and can add kinks only at zeros of g inside affine pieces). An induction on layers yields osc growth at most by a factor m per layer (up to constants), hence $\mathsf{osc}(f) \le Cm^L$. □

Assumption 3 (Linear oscillation creation per block (a proof target))**.** *There exists a regime (e.g. one-sparse/sign-coherent gates and locality) in which the dominant effective preactivation $u^{(\ell)}$ satisfies an* additive *oscillation growth bound*

$$\mathsf{osc}\big(u^{(\ell+1)}(t,\cdot)\big) \ \le \ \mathsf{osc}\big(u^{(\ell)}(t,\cdot)\big) + C_0,$$

with a small constant C_0 (empirically $C_0 \approx 2$ in your plots).

Remark 16 (Why "$2L$" is natural in your block convention)**.** *If one MMNN "block" corresponds (architecturally) to two affine–ReLU stages, then an additive creation of about two new oscillations per block matches the linear pattern "L blocks $\rightsquigarrow$ $2L$ oscillations". Mathematically, this shifts the proof goal from worst-case expressivity (m^L pieces) to a* stability + incremental knot creation *statement along the gradient-flow trajectory.*

What this reduces the training proof to. The expressivity lemma 15 does *not* say that gradient flow will realize the worst-case oscillation count. In your setting, the right claim is instead the linear-growth Assumption 3. Accordingly, the right invariant/proxy to analyze is not a Fourier cutoff: it is a *knot/oscillation creation functional*, e.g. $\mathsf{osc}(u^{(\ell)})$ or a smooth surrogate such as $\mathrm{TV}(\partial_x u^{(\ell)})$. The mean-field task is then to show that, when the residual alternates sign at scale λ^{-1}, the gated dynamics create $O(1)$ additional breakpoints at roughly that scale per block, and that the gain constant in (36) makes the newly created oscillations stable across depth.